ASCII MASTER FLOW — PANEL 1/12: Cycle accounting frame

```
SYSTEM BOUNDARY -> atmosphere, water, soil, sediment and living biomass included
POOL -> amount held in one reservoir at a stated time
FLUX -> transfer between reservoirs per stated time
PROCESS -> biological, chemical or geological transformation
BALANCE -> inputs minus outputs; never assumed from one isolated stock
MUST REMEMBER: Biogeochemical cycles move elements among reservoirs through biological,...
```

ASCII MASTER FLOW — PANEL 2/12: Reservoir classification

```
GASEOUS -> major atmosphere or ocean reservoir
SEDIMENTARY -> major crust, soil or sediment reservoir
CARBON AND NITROGEN -> prominent atmospheric exchange
PHOSPHORUS -> rock-weathering and sediment pathway
CAUTION -> class does not set one universal cycling speed
```

ASCII MASTER FLOW — PANEL 3/12: Carbon movement

```
ATMOSPHERIC OR DISSOLVED CARBON -> photosynthetic fixation
PRODUCER BIOMASS -> consumers and detrital pathways
RESPIRATION -> carbon returned during metabolism
DECOMPOSITION AND COMBUSTION -> additional return pathways
RULE -> stored carbon and annual transfer are different measures
```

ASCII MASTER FLOW — PANEL 4/12: Nitrogen transformation rail

ATMOSPHERIC NITROGEN -> fixation -> ammonia or usable nitrogen forms
ORGANIC NITROGEN -> ammonification -> ammonia
AMMONIA -> nitrification -> nitrite -> nitrate
NITRATE -> plant uptake or denitrification
DENITRIFICATION -> return toward atmospheric nitrogen

ASCII MASTER FLOW — PANEL 5/12: Nitrogen organism map

RHIZOBIUM -> symbiotic fixation associated with legume roots
AZOTOBACTER -> free-living fixation example in the owner
NITROSOMONAS -> ammonia to nitrite step
NITROBACTER -> nitrite to nitrate step
PSEUDOMONAS-TYPE -> denitrification route

ASCII MASTER FLOW — PANEL 6/12: Phosphorus path

PHOSPHATE ROCK -> weathering -> soil or water phosphate
PLANT UPTAKE -> consumer transfer -> detrital return
RUNOFF -> aquatic loading -> sedimentation
MINING AND FERTILISER -> accelerate extraction and dispersal
BOUNDARY -> no significant atmospheric phase in the owner
CLOSE DISTINCTION: Reservoir stock is not flux, residence time is not turnover rate,...

ASCII MASTER FLOW — PANEL 7/12: Cycle-specific human pressure

CARBON -> fossil-fuel combustion and land-cover change
NITROGEN -> synthetic fertiliser and combustion-related reactive nitrogen
PHOSPHORUS -> mining, fertiliser application and runoff
OUTCOMES -> climate forcing, nutrient pollution and resource loss differ
POLICY -> source control must match the cycle and reservoir

ASCII MASTER FLOW — PANEL 8/12: Eutrophication mechanism

NUTRIENT ENRICHMENT -> increased algal or plant production
ORGANIC MATERIAL ACCUMULATES -> decomposer activity rises
OXYGEN DEMAND RISES -> dissolved oxygen may decline
ECOLOGICAL STRESS -> composition changes and mortality may occur
CLASSIFY -> natural ageing versus human-accelerated cultural eutrophication

ASCII MASTER FLOW — PANEL 9/12: Three pyramid parameters

```
NUMBERS -> count of organisms at each trophic level
BIOMASS -> standing living material at a stated time
ENERGY -> transfer through trophic levels over time
SHAPE -> depends on parameter and ecosystem
RULE -> only the energy pyramid is always upright
```

ASCII MASTER FLOW — PANEL 10/12: Aquatic biomass inversion

SMALL PHYTOPLANKTON STANDING CROP -> rapid replacement
HIGH TURNOVER -> continued production over the measurement period
CONSUMER STANDING BIOMASS -> may exceed producer biomass at one instant
ENERGY TRANSFER -> still declines upward
VERDICT -> biomass inversion is not energy inversion

ASCII MASTER FLOW — PANEL 11/12: Heuristic and model firewall

```
ENERGY LOSS -> qualitative thermodynamic direction is robust
TEN-PER-CENT RULE -> owner labels it an average heuristic
OMNIVORY -> weakens a rigid one-level assignment
SEASONALITY -> changes standing stocks and feeding relations
PYRAMID -> useful summary, not a complete food web
```

ASCII MASTER FLOW — PANEL 12/12: PYQ and evidence boundary

ROUTED CONCEPTS -> agricultural nitrogen, rock-weathered phosphorus, fixation links
BASIC PRACTICE -> reservoir, process and organism distinctions
OFFICIAL PAPER -> demand wording only
NO KEY -> no option or answer inferred
LIVE CHECK -> no rate, pool, efficiency or status imported from a stub
EVIDENCE LIMIT: MECHANISM / STATUS / CAUSATION: Name pool, process, direction, limiting...